

Structural Efficiency Cup

Build the highest-scoring structure based on load capacity, mass, and dimensional compliance.

Advanced	2 class periods	Competition
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Engineering Context

A launch provider is selecting a support frame design. Teams will compete using a scoring formula that rewards high load capacity and low structure mass.

What You Will Practice

- Optimize structural geometry
- Use test data to revise a design
- Calculate an efficiency score
- Conduct a fair engineering competition

What You Need

- Standardized structural materials
- Adhesive
- Mass scale
- Load-test fixture
- Official dimension gauge
- Engineering notebook

Student Instructions

- 1 Review the official span, height, and material limits.
- 2 Sketch three possible structural layouts.
- 3 Use a quick model or member-count estimate to select one.
- 4 Build the structure and allow joints to cure.
- 5 Measure mass and verify dimensions before the official test.
- 6 Predict failure location and maximum load.
- 7 Complete the official load test.
- 8 Calculate the competition score using the provided formula.
- 9 Document one redesign for a future round.

Engineering Requirements

- Meet all dimensional limits
- Use only the issued quantity of material
- Structure must be free-standing in the fixture
- Official score uses successful load divided by structure mass
- No changes after official inspection

Constraints & Safety

- Follow destructive-test safety boundary
- Do not stand over the structure
- Use only approved loading increments
- Wear eye protection

What You Submit

- Three concepts
- Inspection record
- Mass and load results
- Efficiency score
- Failure analysis
- Redesign sketch

Reflection Questions

- 1 Which design choice contributed most to the score?
- 2 Where did mass fail to contribute useful strength?
- 3 Did the actual failure match the prediction?
- 4 How could joint consistency be improved?

Engineering Tip

Build and inspect one representative joint early; weak joints can dominate the entire structure.

Common Mistake

Optimizing only for maximum load can produce a heavy structure with a poor efficiency score.

Stretch Goal

Use the same material allowance to create a second structure with a different geometry.

Career Connection

Structural engineers use mass efficiency and margin of safety to compare competing aerospace concepts.